

Part 1: Beginner Exercises (1–20)

Core Skills: Indexing, Basic Printing, Simple Slicing

1. Define a list `fruits = ["apple", "banana", "cherry", "date", "elderberry"]`. Print the first element.
2. Using the `fruits` list, print the last element (use a negative index).
3. Define a tuple `numbers = (10, 20, 30, 40, 50)`. Access and print the 3rd element (index 2).
4. Using `numbers`, print the second-to-last element (negative index).
5. Create a list `colors = ["red", "green", "blue", "yellow", "purple"]`. Print elements at indices 1 and 4 in one line.
6. Try to modify the 2nd element of the `numbers` tuple (index 1) to 25—explain why it fails (write a comment in your code).
7. Define `mixed = [1, "hello", 3.14, True, (6,7)]`. Print the element at index 3.
8. Using `mixed`, print the tuple inside the list (index 4).
9. Slice the `fruits` list to get the first 3 elements; print the result.
10. Slice `numbers` to get elements from index 1 to 3 (inclusive); print the result.
11. Slice `colors` to get the last 2 elements (use negative slicing); print the result.
12. Define `letters = ["a", "b", "c", "d", "e", "f"]`. Slice to get elements from index 2 to the end; print.
13. Slice `letters` to get elements from the start to index 4 (exclusive); print.
14. Print the entire `fruits` list using a slice (no start/end indices).
15. Define `grades = (85, 92, 78, 90, 88)`. Print the average of the first 3 grades (calculate first, then print).
16. Create a list `days = ["Mon", "Tue", "Wed", "Thu", "Fri"]`. Print "Wed" using indexing.
17. Using `days`, slice to get `["Tue", "Wed", "Thu"]`; print the slice.
18. Define `empty_list = []` and `empty_tuple = ()`. Print both, and add a comment explaining if they can be modified.
19. Create a list `nums = [5, 10, 15, 20]`. Print the element at index -2.
20. Using `nums`, slice with step 2 (e.g., `[start:end:step]`) to get `[5, 15]`; print.

Part 2: Intermediate Exercises (21–40)

Core Skills: Reversing, Advanced Slicing, Modification, Combined Operations

21. Define `cars = ["Toyota", "Honda", "Ford", "BMW", "Mercedes"]`. Reverse the list (use `reverse()` method) and print it.
22. Using `cars` (before reversing), create a reversed copy with slicing (`[::-1]`) and print it (don't modify the original list).
23. Define `digits = (1,2,3,4,5,6,7,8,9)`. Reverse the tuple using slicing and print the result (tuples have no `reverse()` method).
24. Create a list `prices = [19.99, 29.99, 9.99, 49.99]`. Slice to get elements from index 1 to the end with step 2; print.
25. Using `prices`, modify the 3rd element (index 2) to 14.99; print the updated list.
26. Define `names = ["Alice", "Bob", "Charlie", "David", "Eve"]`. Print every other name (step 2) starting from index 1.
27. Using `names`, slice to get the last 3 elements in reverse order; print.
28. Create a tuple `coordinates = (100, 200, 300, 400)`. Extract the first two elements into a new tuple `xy`; print `xy`.
29. Define `shopping_list = ["milk", "eggs", "bread", "butter", "cheese"]`. Print the list in reverse order without modifying the original.
30. Using `shopping_list`, remove the 4th element (index 3) and print the updated list (use `pop()`).
31. Create a list `scores = [75, 80, 85, 90, 95]`. Replace the first and last elements with 100; print the updated list.
32. Define `text = ["p", "y", "t", "h", "o", "n"]`. Use slicing to extract "thon" (indices 3–6); print.
33. Using `text`, reverse the list and print the element at index 2.
34. Create a tuple `months = ("Jan", "Feb", "Mar", "Apr", "May")`. Print the tuple in reverse order using slicing.
35. Define `numbers_list = [1,2,3,4,5,6,7,8,9,10]`. Slice to get even-indexed elements (0,2,4...); print.
36. Using `numbers_list`, slice to get odd-valued elements (2,4,6...); print (hint: filter after slicing, or loop).
37. Create a nested list `matrix = [[1,2,3], [4,5,6], [7,8,9]]`. Print the element at row 1, column 2 (value 6).

38. Using matrix, slice to get the second row ([4,5,6]); print.
39. Define tuple_of_lists = ([1,2], [3,4], [5,6]). Modify the first element of the second list (index 1) to 10; print the updated tuple.
40. Create a list words = ["cat", "dog", "bird", "fish"]. Print each element in uppercase (use a loop + upper()).

Part 3: Challenge Exercises (41–50)

Core Skills: Combined Operations, Edge Cases, Problem-Solving

41. Define long_list = list(range(1, 51)) (numbers 1–50). Slice to get numbers from 10 to 20 (inclusive); print.
42. Using long_list, reverse the slice from 10 to 20 and print the result.
43. Create a tuple random = (9, 3, 7, 1, 8, 2, 5). Find the index of the value 8 (use index()); print the index.
44. Define names = ["Anna", "Ben", "Claire", "Dan", "Ella"]. Print the first letter of each name (use indexing on strings inside the list).
45. Using names, create a new list with the names reversed (e.g., "Anna" → "annA"); print the new list.
46. Create a list values = [10, None, 20, None, 30]. Print only the non-None elements (use a loop).
47. Define tuple1 = (1,2,3) and tuple2 = (4,5,6). Concatenate them into a new tuple tuple3; print tuple3.
48. Using tuple3, slice to get the middle 4 elements; print.
49. Create a list temp = [22, 25, 18, 20, 23, 19]. Print the highest temperature (use max()) and its index.
50. Define sentence = ["I", "love", "Python", "programming"]. Join the list into a single string (use join()) and print it (output: "I love Python programming").